

Chapter 9: Foundations of Neural Spaces

One concept, one exercise per slide — practice as you go

Concept: The Perceptron

- Computes a linear score $s = w \cdot x + b$, then applies a hard threshold: predict 1 if $s \geq 0$, else 0
- Rosenblatt's update rule: when misclassified, $w_j \leftarrow w_j + \eta(t - \text{prediction})x_j$
- The update nudges the decision boundary toward the misclassified point



Exercise: A perceptron with $w=[0,0]$, $b=0$ sees the example $x=[1,2]$, target=1. It predicts 0 (since score=0 is not $\geq$ threshold... assume threshold requires strictly positive). Using learning_rate=1, compute the updated weights and bias.

Concept: The Perceptron Convergence Theorem

- IF the training data is linearly separable, the perceptron is GUARANTEED to converge
- Bound: at most R^2/γ^2 updates, where R is the data's radius and γ is the margin
- A smaller margin (harder-to-separate data) means a LARGER guaranteed bound on updates needed



Exercise: Why does the theorem's bound grow without limit as the margin γ shrinks toward zero? What does this predict about training time on nearly-inseparable data?

Concept: The Linear Stacking Collapse Theorem

- Stacking two linear layers with NO activation between them: $W_2(W_1x) = (W_2W_1)x$
- This is just ONE linear layer with combined weights — no extra representational power
- This holds no matter how many linear-only layers you stack — depth alone accomplishes nothing without non-linearity



Exercise: A colleague builds a 10-layer neural network but forgets to add any activation functions between the layers. How many DISTINCT linear regions can this network actually represent, no matter how it's trained?

Concept: Why the Perceptron Cannot Learn XOR

- XOR: $(0,0) \rightarrow 0$, $(0,1) \rightarrow 1$, $(1,0) \rightarrow 1$, $(1,1) \rightarrow 0$ — the positive and negative classes are NOT linearly separable
- No single straight line can separate the two classes
- Minsky and Papert's 1969 proof of this fact triggered the first "AI winter"



Exercise: Plot the four XOR points mentally on a 2D grid. Explain in one sentence why no single straight line can separate the two classes.

Concept: Escaping XOR — The Hidden Layer

- Add ONE hidden layer with a non-linear activation between the layers
- The hidden layer "folds" the input space so the previously inseparable classes become separable
- A hand-built 2-neuron ReLU hidden layer can solve XOR exactly, with no training at all — just the right weights



Exercise: Two ReLU hidden units compute $h1 = \text{ReLU}(x1+x2)$ and $h2 = \text{ReLU}(x1+x2-1)$. The output is $h1 - 2 \cdot h2$. Evaluate this for input (1,1) and check it matches XOR's expected output of 0.

Concept: Sigmoid and Tanh — Derivatives and Saturation

- Sigmoid: $\sigma(z) = 1/(1+e^{-z})$, derivative $\sigma'(z) = \sigma(z)(1-\sigma(z))$, max value 0.25 at $z=0$
- Tanh: derivative $1 - \tanh^2(z)$, max value 1 at $z=0$
- Both derivatives approach exactly ZERO as $|z|$ grows large — this is "saturation"



Exercise: A neuron's pre-activation value is $z=15$ under a sigmoid activation. Is this neuron's gradient likely to be useful for learning, or effectively dead? Why?

Concept: The Vanishing Gradient Problem

- Backpropagating through many saturating (sigmoid/tanh) layers multiplies many small derivatives together
- Sigmoid's derivative maxes out at 0.25 — through 10 layers, the gradient can shrink by a factor of up to 0.25^{10}
- This geometric decay made networks deeper than a couple of layers effectively untrainable for two decades



Exercise: If every layer's sigmoid derivative happens to be at its maximum value of 0.25, roughly what fraction of the original gradient survives after passing back through 5 layers?

Concept: Why ReLU Breaks the Vanishing Gradient

- $\text{ReLU}(z) = \max(0, z)$ — derivative is EXACTLY 1 for $z > 0$, not a shrinking fraction
- Stacking many ReLU layers does NOT geometrically shrink the gradient the way sigmoid does
- Trade-off: ReLU's derivative is exactly 0 for $z \leq 0$ — a "dying ReLU" can get permanently stuck



Exercise: Explain why ReLU solves the vanishing-gradient problem for POSITIVE pre-activations, but introduces a different failure mode for NEGATIVE ones.

Mastery Check: Are You Ready to Move On?

If you can answer every question below confidently, you have mastered Chapter 9.

1. Write the Rosenblatt perceptron update rule and apply it by hand to one misclassified example.
2. State the Perceptron Convergence Theorem and explain what R and γ represent.
3. Prove (in outline) the Linear Stacking Collapse Theorem for a zero-bias two-layer network.
4. Explain why XOR is not linearly separable and why this defeated the original perceptron.
5. Explain how adding a hidden layer with a non-linear activation solves XOR.
6. Derive the sigmoid derivative and state its maximum value and where it occurs.
7. Explain the vanishing gradient problem as a consequence of the chain rule and saturating derivatives.
8. Explain why ReLU avoids vanishing gradients on its positive side, and describe the 'dying ReLU' failure mode.
9. Explain why a bias term is necessary for a perceptron to represent decision boundaries not through the origin.
10. Implement a perceptron from scratch and empirically verify it converges on AND/OR but not on XOR.